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Dynamics of an elliptical cylinder in confined Poiseuille flow

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ABSTRACT

Flows of solid particles in suspension are ubiquitous in both nature and industry. Compared to a spherical particle, the dynamics of a non-spherical particle in flow is much less understood, especially its interaction with a micro-confined environment. We consider an elliptical particle because its different aspect ratios can represent a large family of non-spherical shapes. To capture the complex dynamic interface between the particle and the flow, we employ the smoothed particle hydrodynamics method and benefit from its Lagrangian property. In particular, we consider an elliptical cylinder in confined Poiseuille flow and systematically study the effects of five factors: the confinement strengths, the particle Reynolds numbers between 0.1 and 10, particle initial positions/orientations, and the particle aspect ratios, respectively. We identify three types of periodic motion at steady state and they are tumbling, oscillation with either major or mini axis along the flow. In weakly confined channels, the particle always tumbles and has determined focusing positions off the centerlines, which depend mainly on the competition between the shear gradient lift and wall-induced force in the transverse direction. In strongly confined channels, the particle has steady oscillations at the centerlines, and its actual state depends on the Reynolds number, initial states, and aspect ratios of the particle. Our study provides a valuable insight into the dynamics of non-spherical particles in microfluidic systems.

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I. INTRODUCTION

The flow of solid particles within a microchannel is a typical scenario in the context of micro-/nano-fluidics, biomedical applications, and industrial processing.[1,2] For example, in pipelines, particle settling has a significant impact on the transport and filtration;[3–5] in microcirculation, the transport of solid particles in blood flow is fundamental to understanding the mechanism of drug delivery.[6,7] In particular, the phenomenon of inertial focusing is of great importance in characterizing the flow behavior of suspensions and has been widely used to separate particles in microfluidic systems.[8–13]

In pipe flow, the projections of individual spherical particles form a ring on the cross section of the pipe at steady state, irrespective of their initial positions. The radius of the ring is about $0.6R$, where R is the radius of the pipe. This phenomenon was first observed experimentally by Segré and Silberberg[14] and is now known as inertial focusing. Among other early analytical work, Ho and Leal employed the method of reflection to study a neutrally buoyant cylinder in two-dimensional (2D) Couette and Poiseuille flows.[15] By taking into account the inertial effects, their results are in excellent agreement with the experiments. For a non-spherical particle in a flow, it is formidable to take account of the nonlinear effects, so most analytical work has been based on Stokes flows. This dates back to the classic work by Jeffery, who focused on an ellipsoid under shear flow and derived its corresponding orbit.[16] Chwang also made an important extension in an unbounded quadratic flow.[17] Other non-spherical particles, such as the slender body and thin disk in Stokes flow, have also been studied analytically.[18,19] Since Stokes flows are time-reversible, there is no net migration of the particle across the transverse direction.

With the advancement of computational fluid dynamics (CFD) methods over the last few decades, particle/fluid inertia, non-spherical shapes, non-neutral buoyancy, or wall confinement is no longer infeasible. To be more precise about the different equations solved by different CFD techniques, we give a brief classification of the models to facilitate the discussion.

Model I. It begins with the Stokes equations as follows:

$$\nabla \cdot \mathbf{v} = 0, \tag{1a}$$

$$\nabla p = \mu \nabla^2 \mathbf{v}, \tag{1b}$$

$$\frac{d\mathbf{V}}{dt} = 0, \quad \frac{d\mathbf{\Omega}}{dt} = 0, \tag{1c}$$

where $\mathbf{v}$, p, and μ are the velocity, pressure, and viscosity of the fluid, respectively. Equation (1a) is the continuity equation for an incompressible flow, while Eq. (1b) is the momentum balance between the pressure and viscous forces. Given the no-slip boundary condition at interfaces such as the particle surface and the enclosing walls, the flow can be solved numerically with an excellent accuracy.[1] Equation (1c) describes the motion of an inertial-less particle, which instantaneously adjusts its translational velocity $\mathbf{V}$ and rotational velocity $\boldsymbol{\Omega}$ according to the flow so that there is no net force or torque on the particle.

Model II. If the particle has inertia, its momentum equations become

$$M\frac{d\mathbf{V}}{dt} = \mathbf{F}, \quad \mathbf{I}\frac{d\boldsymbol{\Omega}}{dt} = \mathbf{T}, \tag{2}$$

where M and $\mathbf{I}$ are mass and moment of inertia, respectively. First, the flow equations are solved, and then the force and torque of the particle are computed as integrals of the stress σ at the surface $\mathbf{F} = \int \sigma dS$ and $\mathbf{T} = \int \sigma \times \mathbf{X} dS$. Thereafter, the velocity and position of the particle are integrated by one time step. With the updated state of the interface, the flow equations are solved again for a new steady state. This process is repeated in time.

Model III. If the flow takes time to evolve, while the nonlinear effects are still negligible, the momentum equations change to

$$\rho\frac{\partial \mathbf{v}}{\partial t} = -\nabla p + \mu\nabla^2\mathbf{v}. \tag{3}$$

The flow fields become time-dependent, and the particle dynamics remain the same as in model II.

Model IV. When both the transient and nonlinear effects of the flow are important, the momentum equations take the form

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}\cdot\nabla\mathbf{v}\right) = -\nabla p + \mu\nabla^2\mathbf{v}. \tag{4}$$

Equation (4) is known as the Navier–Stokes (NS) equations, and the particle dynamics remain unchanged as in models II and III. Note that model IV is the most expressive model, and it degenerates into other models given appropriate control parameters.

Sugihara-Seki[20,21] employed the finite element method with a static grid to solve model I for an elliptical/ellipsoidal particle in Poiseuille flow and also considered the effects of confining walls. The trajectory of the particle was constructed by connecting adjacent solutions at quasi-steady states. Three modes of motion for the particle were then identified: tumbling between the channel centerline and the wall; oscillation near the centerline with the minor axis along the flow; and oscillation near the wall with the major axis along the flow. Using a similar methodology but a dynamic procedure of frequent grid generation, Feng *et al.*[22] solved both models II and III to emphasize the importance of the particle inertia and/or flow unsteady term. The motivation was due to the fact that the unsteady particle and flow terms approach zero as $Re^{-1/2}$ in Poiseuille flow, which is slower than Re of the nonlinear term.[22] Here, Re is the Reynolds number defined as the ratio of inertial and viscous forces, using the size of the particle as the length scale. They found that the dynamics of the particle described by model II is similar to those of model I, while there is no steady oscillation of the particle near the wall in model III.

Model IV has been studied more extensively. In particular, the lattice Boltzmann method appears to be the most popular method for dealing with the particle suspensions described by this model. The applications include sedimentation of an elliptical particle in a two-dimensional tube,[23] inertial migration of an oblate particle in a square pipe,[5] dynamics of spheroids in Poiseuille flow considering the influence of intermediate Reynolds numbers and particle aspect ratios,[24] inertial focusing behavior of spheroids in square ducts,[12] and migration and rheotaxis of elliptical squirmers in a Poiseuille flow,[11] among others. In addition, Lagrange multiplier-based fictitious domain methods have also been widely used to study the motion of neutrally buoyant ellipsoids in Couette/Poiseuille flows with different Reynolds numbers, particle aspect ratios, and solid area fractions.[25–27] Nevertheless, the dynamics of a non-spherical particle in flow within a confined channel described by model IV is not adequately studied.

Smoothed particle hydrodynamics (SPH) was originally developed for the study of astrophysics. Since then, it has been largely extended and widely applied to various fluid dynamics problems.[28,29] In particular, due to its Lagrangian nature, SPH is advantageous for dealing with complex dynamic interface and has previously been employed to solve model IV to study hydrodynamic interactions between multiple spherical particles and also with confining walls.[30–33] In this study, we apply SPH to systematically study the dynamics of an elliptical cylinder in Poiseuille flow and its dependence on five factors: confinement strengths, Reynolds numbers (ranging from 0.1 to 10), initial states including position and orientation, and particle aspect ratios. We found that in weakly confined channels, the particle always tumbles and has a determined inertial focusing position; in strongly confined channels, the particle has two types of steady oscillation around the centerline, and its actual steady state depends on the other four factors.

The rest of the work is organized as follows. In Sec. II, we introduce the numerical method. We present our results and discussions in Sec. III. Finally, we summarize this work in Sec. IV.

II. THE METHOD

We employ the SPH method to numerically solve model IV, while accurately capturing the no-slip boundaries at the particle surface and the solid walls. Since the interface between the elliptical particle and the fluid is complex in motion, we adopt a special treatment for the boundary condition from previous work.[34,35]

A. The SPH method

We take a weakly compressible formulation to resemble the incompressible flows, and the continuity equation reads

$$\frac{d\rho}{dt} = -\rho\nabla\cdot\mathbf{v}, \tag{5}$$

which replaces Eq. (1a). An equation of state relating the pressure to the density is given as

$$p = c^2(\rho - \rho_0) + \chi, \tag{6}$$

where ρ_0 is the equilibrium density, and χ is a positive constant for enforcing the non-negativity of pressure on each discrete particle. An artificial sound velocity c is chosen based on a scale analysis,[36] and its actual value depends on the flow characteristics so that the density variation is negligible.

The continuity equation is discretized as[28]

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$$\frac{d\rho_i}{dt} = \rho_i \sum_j \frac{m}{\rho_j} \mathbf{v}_{ij} \cdot \frac{\partial W}{\partial r_{ij}} \mathbf{e}_{ij}, \tag{7}$$

where ρ_j is the density for particle j, and m is the identical mass for each particle; $\mathbf{r}_{ij}$ and $\mathbf{v}_{ij}$ represent the relative position and velocity of particles i and j, respectively. Furthermore, $r_{ij} = |\mathbf{r}_{ij}|$ and $\mathbf{e}_{ij} = \mathbf{r}_{ij}/r_{ij}$. We adopt the quintic spline as the kernel function $W(r_{ij})$,[36] which vanishes beyond a cutoff radius r_c. After several time steps, numerical errors in the density integration can accumulate, so we restore the density every ten time steps via the summation equation

$$\rho_i = \sum_j m_j W_{ij}. \tag{8}$$

We note that the results obtained in this way are consistent with those obtained by restoring the density at each step, but the former significantly reduces the computational effort.

To further minimize numerical errors due to irregular distributions of SPH particles, we adopt the transport-velocity formulation, and the discrete momentum equation reads[37]

$$\frac{\tilde{d}\mathbf{v}_i}{dt} = \frac{1}{m_i} \sum_j \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right) [\mathbf{f}_p + \mathbf{f}_\mu + \mathbf{f}_A] + \mathbf{g}_i. \tag{9}$$

Here, $\tilde{d}(*)/dt$ represents a new definition of the material derivative. σ is the number density defined as the ratio of ρ and mass m. $\mathbf{f}_p$, $\mathbf{f}_\mu$, and $\mathbf{f}_A$ are the pressure term, the viscous term, and the additional term due to convection as follows:[37]

$$\mathbf{f}_p = -\frac{\rho_j p_i + \rho_i p_j}{\rho_i + \rho_j} \frac{\partial W}{\partial r_{ij}} \mathbf{e}_{ij}, \tag{10}$$

$$\mathbf{f}_\mu = \frac{2\mu_i \mu_j}{\mu_i + \mu_j} \frac{\mathbf{v}_{ij}}{r_{ij}} \frac{\partial W}{\partial r_{ij}}, \tag{11}$$

$$\mathbf{f}_A = \frac{1}{2} (\mathbf{A}_i + \mathbf{A}_j) \cdot \frac{\partial W}{\partial r_{ij}} \mathbf{e}_{ij}. \tag{12}$$

Here, we use the density-weighted pressure and interparticle-averaged shear viscosity. $\mathbf{A} = \rho \mathbf{v}(\tilde{\mathbf{v}} - \mathbf{v})$ is a tensor from the dyadic product of the two vectors. Furthermore, $\mathbf{v}$ is the velocity for momentum and force calculations, while $\tilde{\mathbf{v}}$ is the modified transport velocity for updating the position of each fluid particle. The discrete form of $\tilde{\mathbf{v}}$ is calculated as

$$\tilde{\mathbf{v}}_i = \mathbf{v}_i(t) + \delta t \left(\frac{\tilde{d}\mathbf{v}_i}{dt} - \frac{\chi}{m_i} \sum_j \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right) \frac{\partial W}{\partial r_{ij}} \mathbf{e}_{ij} \right), \tag{13}$$

where δt is the time step, and the positive constant χ only appears here but not in the momentum equation. The momentum and position of each particle are integrated by velocity Verlet method, and the time step is restricted due to sonic speed, viscosity, and body force.[36]

B. Boundary treatment at fluid–solid interface

We use SPH boundary particles to represent any solid body, which provide the flexibility to describe arbitrary boundaries, as depicted in Fig. 1. These boundary particles are initialized in the same way as the fluid particles, with the fluid–solid boundaries determined before the simulation. In particular, they have identical mass and resolution Δx to the fluid particles, ensuring that a fluid particle near the boundary has a complete support domain.

To implement the no-slip boundary condition at the interface, we assign to each boundary particle i an artificial velocity, which is determined by the properties of a virtual particle E and a prescribed slip length at the interface.[35] The slip length is set to zero in this work, which represents the no-slip condition. The position $\mathbf{x}_E$ and velocity $\mathbf{v}_E$ of the virtual particle E are calculated using SPH interpolations,[35] which represent an average effect due to neighboring fluid particles on the boundary particle i, as shown in Fig. 1. Thereafter, an artificial velocity $\mathbf{v}_i = (v_i^\tau, v_i^n)$ is assigned to the particle i as

$$v_i^\tau = -\frac{d_i}{d_E} v_E^\tau + \frac{d_E + d_i}{d_E} v_G^\tau, \tag{14}$$

$$v_i^n = -\frac{d_i}{d_E} (v_E^n - v_G^n) + v_G^n. \tag{15}$$

Here, τ and n represent the tangential and normal directions at the interface plane, respectively. d_i and d_E are the distances of the boundary particle i and the virtual particle E from the interface, respectively. $\mathbf{v}_G = (v_G^\tau, v_G^n)$ is the velocity of the intersection point at the interface, which is available from the motion of the rigid body. Note that the artificial velocity of the particle i is only used in the calculation of the viscous force in Eq. (9), but is not intended for the kinematics of the solid. The density ρ_i of particle i is calculated in a similar way to satisfy the pressure boundary condition at the interface, and we refer to Ref. 35 for details.

Given properly implemented boundary conditions, we can calculate the total force $\mathbf{F}_c$ and torque $\mathbf{T}_c$ on the moving rigid body. These are the sum of the forces of all the SPH boundary particles that make up the rigid body, as follows:

$$\mathbf{F}_c = \sum_i \mathbf{f}_i = \sum_i (\mathbf{f}_p + \mathbf{f}_\mu + \mathbf{f}_A), \tag{16}$$

$$\mathbf{T}_c = \sum_i (\mathbf{r}_i - \mathbf{X}_c) \times (\mathbf{f}_p + \mathbf{f}_\mu + \mathbf{f}_A), \tag{17}$$

where $\mathbf{X}_c$ is the center of mass (CoM) of the rigid body, and $\mathbf{r}_i$ is the coordinate position vector of the particle. Each $\mathbf{f}_i$ is calculated as the sum of its interactions with the fluid particles in its support domain.

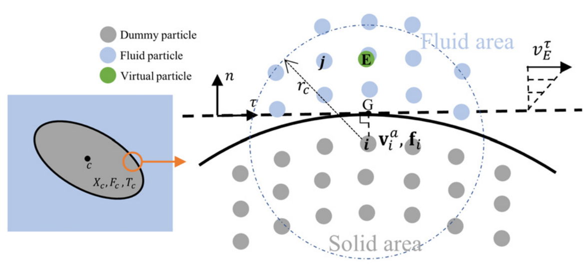

FIG. 1. Schematic of a moving rigid ellipse and distribution of SPH particles for the boundary treatment. The ellipse has its center of mass at $\mathbf{X}_c$. Consider a SPH boundary particle i inside the ellipse, and its tangent plane intersects the interface at G. A virtual particle denoted by E is generated by SPH summations, which represents the average effect of each neighboring fluid particle j on the boundary particle i. An artificial velocity $\mathbf{v}_i^a$ on the boundary particle i is calculated based on the velocity $\mathbf{v}_E = (v_E^\tau, v_E^n)$ of the particle E, while taking into account distances of particles E and i to the interface.

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Thereafter, we can update the linear velocity $\mathbf{V}_c$, the angular velocity $\boldsymbol{\Omega}_c$, and the position $\mathbf{X}_c$ of the rigid body. Finally, we can update the positions of the SPH boundary particles inside the rigid body according to the motion of the rigid body. More information about the dynamics of the rigid body can be found in Appendix A.

C. Validation

To assess the validity of the method, we perform SPH simulations of an elliptical cylinder in shear flow at low Reynolds number and compare the results with theoretical solution. The motion of an ellipse in shear flow is a degeneration of a three-dimensional ellipsoid.[16] Hence, the angular velocity Ω and the rotational angle θ of the ellipse are given by

$$\Omega = \frac{\dot{\gamma}}{a^2 + b^2}\left(a^2\cos^2\theta + b^2\sin^2\theta\right), \tag{18}$$

$$\tan\theta = \frac{a}{b}\tan\frac{ab\dot{\gamma}t}{a^2 + b^2}, \tag{19}$$

where a and b are the major axis and the minor axis of the ellipse, respectively. $\dot{\gamma}$ is the shear rate. The shear flow is generated by two parallel walls moving in opposite directions at a constant velocity $V_0 = 0.05$, and the flow direction has periodic boundaries. We take $2a = 4b = 1$ and set the resolution of the SPH particles to be $\Delta x = b/20$ and $r_c = 3\Delta x$, after preliminary resolution studies. To reduce the effects of the walls and periodic boundaries, the computational domain is taken as a large square box with length $L = 20a = 10$, and the ellipse is initially placed in the center of the box. Furthermore, the density ρ and kinematic viscosity ν of the fluid are set to 1. Therefore, the shear rate is $\dot{\gamma} = 2V_0/L = 0.01$, and the corresponding Reynolds number $Re = 4a^2\dot{\gamma}/\nu = 0.01$ is sufficiently small. Here, $\nu = \mu/\rho$ is the kinematic viscosity. The results of the simulations are shown in Fig. 2 and are in excellent agreement with the theoretical solution.

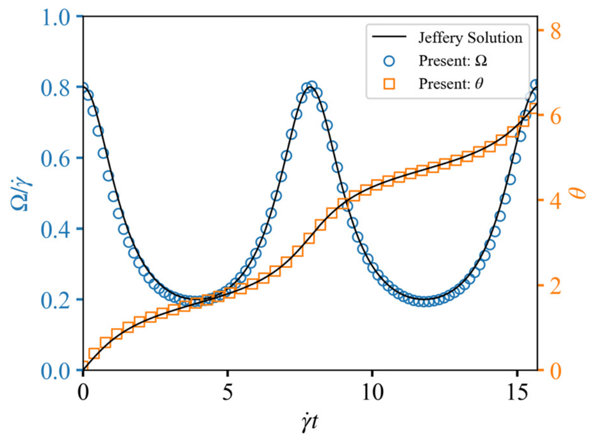

FIG. 2. Comparison between SPH simulation and Jeffery orbit for an ellipse in shear flow. Angular velocity and time are dimensionless. The rotational angle is expressed in radians.

III. RESULTS AND DISCUSSION

A body force $\mathbf{f}_b$ per unit mass is applied along the periodic direction x to drive the flow, and two parallel walls confine the flow in the y-direction. The mid-channel is situated at $y = 0$, and the half-width of the channel is d. In the absence of the ellipse, this is the plane Poiseuille flow, and the velocity profile is parabolic with the maximum value U_{max} at the centerline: $U_{max} = f_b d^2/2\nu$.

An elliptical cylinder with semi-axes a and b is suspended in the flow with neutrally buoyance as shown in Fig. 3. Its shape is defined by the aspect ratio $\alpha = a : b$, while the channel–particle size ratio is defined by $\beta = d : a$. The center of mass of the ellipse is denoted by (X, Y), and the angle between the semi-major axis and the x-axis is denoted by θ. Since the flow is periodic in the x-direction, we are not concerned with the specific location in the x-direction and, therefore, assign $X = 0$ to be the initial location of the ellipse. The density ρ and the kinematic viscosity ν are set to 1 so that the Reynolds number is defined as

$$Re = \frac{2U_{max}a}{\nu} = \frac{f_b d^2 a}{\nu^2}. \tag{20}$$

As we keep a and ν constant, we obtain four different particle Reynolds numbers $Re = 0.1, 0.5, 1.0, \text{and} 10$ by adjusting the body force $\mathbf{f}_b$ according to different values of d. To eliminate any effects due to the periodic boundaries, we set the channel sufficiently long with $L = 10a$.

We adopt a resolution of 20 SPH particles along the minor semi-axis of the ellipse, that is, $\Delta x = b/20$, unless stated otherwise. This resolution is considered sufficient to provide sufficient SPH particles in the gap between the ellipse and the wall to accurately capture the lubricating effects.[38] Selective simulations with higher resolutions are presented in Appendix B, which confirms that this choice of resolution delivers convergent/physics results.

The focusing position of a particle in a Poiseuille flow is primarily a consequence of the balance between wall-induced interaction force and shear gradient lift.[9] The former arises from squeezing the fluid between the particle and the wall, which generates a repulsive force directed away from the wall.[39] The magnitude of this force increases as the distance between the particle and the wall decreases and is also related to the Reynolds number.[40] The second force is mainly attributed to the shear gradient due to the curvature of the undisturbed velocity profile. The velocity profile of the fluid is asymmetric with respect to that of the particle, inducing a force toward the wall,[8] which in turn depends also on the Reynolds number. Another type of force is the Saffman lift due to the difference in velocity between the particle and the fluid,[41] which pushes the particle toward the higher velocity region of the fluid. Particle rotation also leads to an additional lift known as the Magnus force, initially discovered by Rubinow and

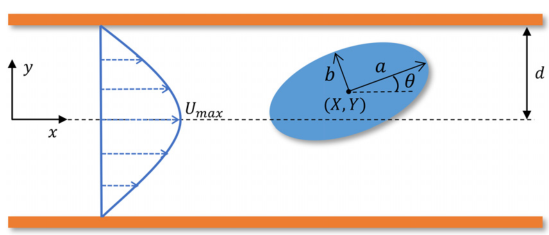

FIG. 3. The geometry of an elliptical cylinder with neutral buoyancy in Poiseuille flow. a and b are the major and minor semi-axes of the ellipse, respectively. θ is the angle between the major axis and the x-direction. d is the half-width of the channel.

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Keller.[42] Both the Saffman and the Magnus forces have a negligible effect on the lateral migration of the particle.[15,43] Therefore, it is reasonable to focus on the competition between the wall-induced force and the shear gradient lift force.

We will first consider the dynamics of an ellipse with aspect ratio of $\alpha = 2$ and study the influence of different values of β and Re on the focusing position, steady orientation, and types of motion. We further study how an initial configuration (Y_0, θ_0) affects the steady motion. Moreover, we analyze the dynamics of a more slender ellipse with $\alpha = 3$.

A. Focusing positions for particle aspect ratio $\alpha = a : b = 2 : 1$

We start with a particle aspect ratio of $\alpha = 2$ and examine the effects of different channel–particle size ratios β and Reynolds numbers Re. We select ten values of β: 0.9, 1.1, 1.2, 1.4, 1.6, 1.8, 1.9, 2.0, 2.4, and 3.0 and four values of Re: 0.1, 0.5, 1.0, and 10 to perform 40 groups of simulations. Moreover, each simulation in every group starts with a different initial position and/or orientation of the particle, effects of which will be discussed later.

We present the equilibrium positions for the center of mass (CoM) of the ellipse in Fig. 4, where the vertical bars indicate the amplitudes of oscillations in the y-direction at steady states.

When the Reynolds number is as low as $Re = 0.1$, the ellipse exhibits a consistent behavior of moving toward the centerline of the channel *regardless of its initial states* due to the wall-induced interaction force. For the channel–particle size ratio β up to 3.0, the shear gradient lift force is still not comparable to the wall-induced force and therefore unable to push the particle away from the centerline. We speculate that the particle may shift away from the centerline for a very large β, which is beyond the focus of this work. Although the Reynolds number is very small, the results are fundamentally different from Sugihara-Seki's work,[20,21] where the particle in Stokes flow does not drift in the transverse direction at all.

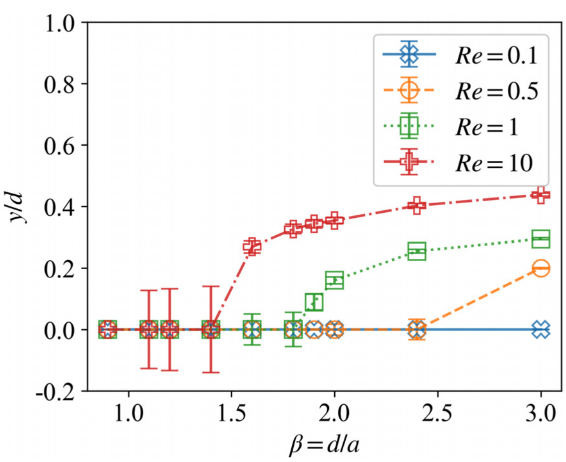

FIG. 4. Focusing positions of the center of mass of the ellipse with $\alpha = 2$ for different channel–particle size ratios β and Reynolds numbers Re at steady states. The vertical bars indicate the amplitude of the position oscillations in the y-direction at steady state.

As the Reynolds number increases, the shear gradient lift force becomes stronger so that the equilibrium position of the particle shifts away from the centerline for β above a certain value. At $Re = 0.5$, this takes place for $\beta \geq 3.0$; at $Re = 1.0$, for $\beta \geq 1.9$; and at $Re = 10$, for $\beta \geq 1.6$: with a clear trend that a larger inertia starts to drift the particle away from the centerline at a smaller β. The off-centerline equilibrium positions represent the typical inertial focusing phenomena. On the other hand, for the channel–particle size ratio below a certain value, the shear gradient lift force cannot outcompete the wall-induced force so that the particle stays on the centerline at steady states. At $Re = 0.5$, this happens for $\beta \leq 2.4$; at $Re = 1.0$, for $\beta \leq 1.8$; and at $Re = 10$, for $\beta \leq 1.4$.

We note that the geometry of the elliptical particle is anisotropic that causes the equilibrium position to oscillate in the transverse direction. The oscillation is more pronounced for a stronger confinement (smaller β) and a larger Reynolds number, as indicated by the vertical bars on Fig. 4. This has to do with its orientation and types of steady motion, which will be explained next.

B. Orientations and types of periodic motion at steady state

At steady states, we observe three types of periodic motion of the ellipse: (1) major axis along the flow as it oscillates around $\theta = 0°$, or simply called "lying" according to the coordinate; (2) minor axis along the flow as it oscillates around $\theta = 90°$, or simply called as "standing"; and (3) tumbling motion, as sketched in Fig. 5.

According to different β and Re, we summarize the steady behaviors by a 2D phase diagram in Fig. 6, where we can identify four regions on the diagram. Region I represents a strong confinement with $\beta = 0.9$. Since $\beta < 1$, there is geometrically no space for the ellipse to "stand up." Therefore, the ellipse lies down around the centerline with its major axis oscillating along the flow, regardless of the Reynolds number.

In region II of β being slightly larger than 1, the ellipse always exhibits one of the two types of oscillations around the centerline:

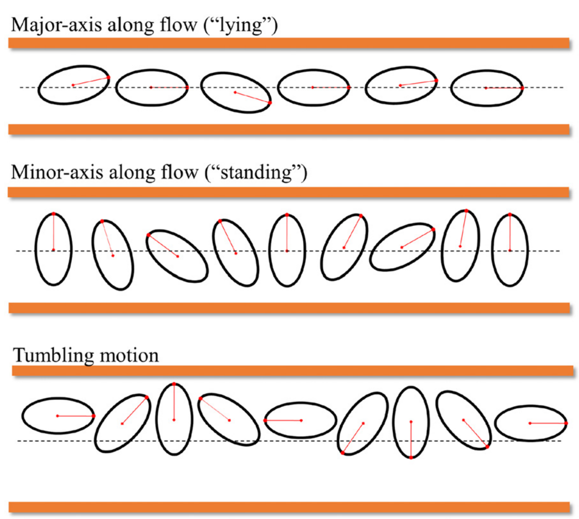

FIG. 5. Schematic of the three types of periodic motion at steady states.

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either major or minor axis along the flow (lying or standing). The specific steady state depends on the particle's initial position and orientation, and we summarize the dependence by a few representative initial states in Table I. The ellipse exhibits steady standing motion type when the initial angle $\theta_0 \geq 18°$, regardless of Re and Y_0; when the initial angle $\theta_0 < 18°$, its steady motion further depends on Re and/or Y_0. For example, at $\theta_0 \leq 3°$ and $Re = 1$, the ellipse remains lying at steady state; at $\theta_0 \leq 3°$ and $Re = 10$, the ellipse prefers lying for smaller Y_0 (= 0.02), whereas prefers standing for larger Y_0 (= 0.2) at steady state. The results for $Re = 0.1$ and $Re = 0.5$ mirror those for $Re = 1$ and therefore are omitted. The underlying cause for the emergence of these two distinct motion types is attributed to the relatively small channel–particle size ratio, which implies a strong confinement strength.

In region III of relative less confinement/larger β, the particle uniquely prefers to oscillate around the centerline with its minor axis along the flow (standing), regardless of its initial states. If the Reynolds number is large enough, the shear gradient lift grows stronger and pushes the particle away from the centerline to new focusing positions as described in region IV. Meanwhile, the particle tumbles at steady state regardless of its initial states.

To summarize this part, focusing positions on the centerline correspond to standing or lying motion type; focusing positions off the centerline (that is, the inertial focusing phenomena) always correspond to tumbling motion type. With this phase diagram, we can predict some other types of periodic motion at different (β, Re). For example, on the top-right side of region IV, tumbling should take place, which have been extensively studied in the past.[8–10,24]

C. Dynamics in one period at steady state

Three types of periodic behaviors for one full period at steady state are shown in Fig. 7, where we choose typical results at $Re = 10$

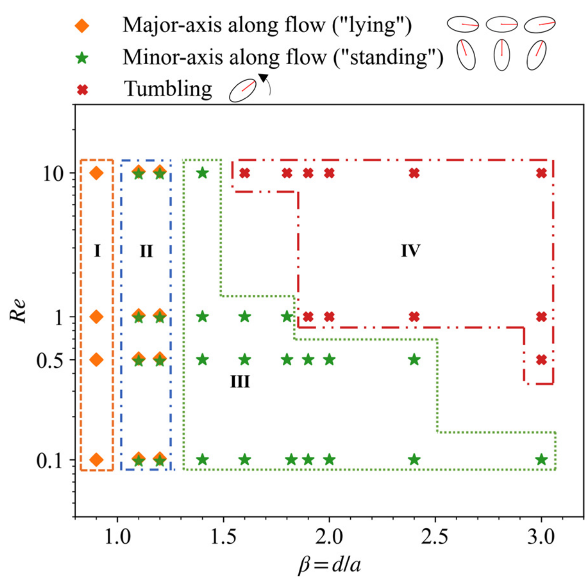

FIG. 6. Phase diagram for the steady behaviors of the ellipse with $\alpha = 2$ with respect to the channel–particle size ratio β and Reynolds number Re.

TABLE I. Relations between the initial position (Y_0/d)/orientation (θ_0) and the steady behavior in region II on Fig. 6. "S" stands for standing motion and "L" for lying motion at steady states. "-" means geometrically not applicable.

Re	β	$Y_0/d\backslash\theta_0$	0°	3°	9°	18°	87°
Re = 1	1.1	0.2	L	L	L	S	-
		0.02	L	L	L	S	S
	1.2	0.2	L	L	L	S	-
		0.02	L	L	S	S	S
Re = 10	1.1	0.2	S	S	S	S	-
		0.02	L	L	S	S	S
	1.2	0.2	S	S	S	S	-
		0.02	L	L	S	S	S

and $Re = 1$ for illustration, including the lying and standing motions for $\beta = 1.2$ and the tumbling motion for $\beta = 1.6$. The horizontal coordinate indicates a full period, and 0 is the start of the period. Note that different motions have different periods, which are described in the caption.

For the lying motion, the ellipse experiences shear force of the flow that induces rotation due to the velocity gradient and curvature. However, the wall pressure limits this rotation, with any slight alteration in θ promptly being restrained by the wall. Consequently, the major axis of the ellipse oscillates along the flow direction with a very slight amplitude, and the ellipse weaves back and forth in the lateral direction with a small magnitude, as shown in Fig. 7(a).

For the standing motion, two different trajectories of CoM in the lateral direction are presented in Fig. 7(b) for $Re = 10$ and Fig. 7(c) for $Re = 1$. In the former case, when the ellipse turns through $\theta = 90°$, there are three turning points [corresponding to points B, C, and D in Fig. 7(b)], and the velocity in the y-direction changes drastically. The trajectory from A to E resembles the letter "M." In the latter case, there is only one turning point [corresponding to point B in Fig. 7(c)], and the velocity in the y-direction is mild. The entire motion is smooth, and the trajectory resembles that of an arch bridge. We observe that the "M-shaped" trajectory occurs exclusively at a Reynolds number of 10 for the standing type. This phenomenon is attributed to the significant shear gradient lift experienced by the ellipse, causing it to approach the wall and consequently endure an increased wall pressure.

For the tumbling motion, the ellipse has a minimum longitudinal velocity when it rotates to 90° and a maximum at 0°; the lateral displacement of the ellipse changes abruptly when it rotates past 90°, increasing and then decreasing violently until it reaches a minimum at 90°. This process repeats itself. The lateral velocity also undergoes a similar abrupt change. This phenomenon is similar to the tumbling motion obtained by Sugihara-Seki[20] under Stokes flow, except that Seki's result is symmetrical before and after the ellipse crosses 90°, whereas ours is not symmetrical due to the presence of additional inertial forces. The ellipse has its maximum angular velocity at 90° and its minimum at 0°. This periodic evolution of the angular velocity, similar to that of the delta function, is consistent with the results of Chwang's angular velocity of the ellipsoid under unbounded conditions as well as Sugihara-Seki's two-dimensional Stokes flow.

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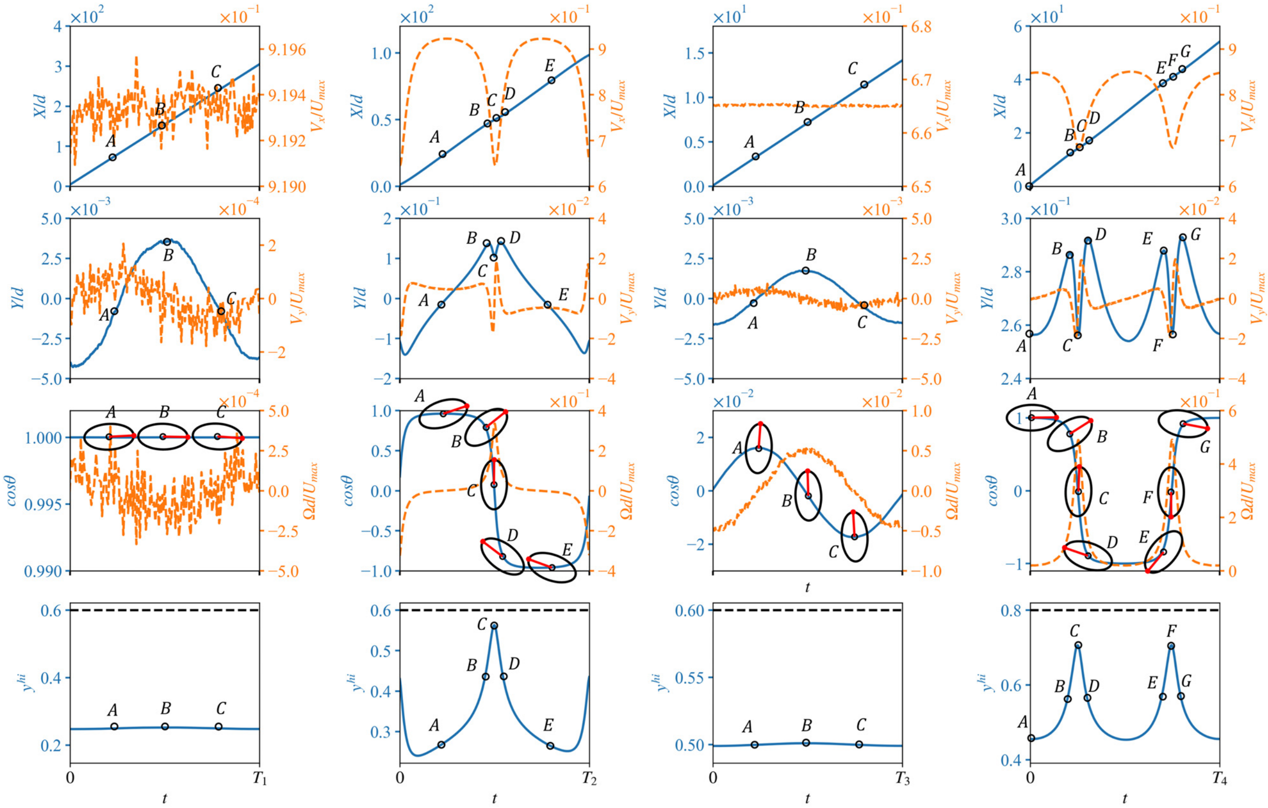

(a)Lying at $\beta = 1.2, Re = 10$ (b)Standing at $\beta = 1.2, Re = 10$ (c)Standing at $\beta = 1.2, Re = 1$ (d)Tumbling at $\beta = 1.6, Re = 10$

FIG. 7. Four examples are chosen to demonstrate the different types of periodic behaviors for one full period at steady state. Horizontal coordinate represents one period, which has different values ($T_1 - T_4$) for the four cases. (a) The lying motion at $\beta = 1.2$ and $Re = 10$: $T_1 = 19.6$ and $-0.19^\circ \le \theta \le 0.19^\circ$, (b) the standing motion at $\beta = 1.2$ and $Re = 10$: $T_2 = 7.0$ and $16^\circ \le \theta \le 164^\circ$, (c) the standing motion at $\beta = 1.2$ and $Re = 1$: $T_3 = 11.1$ and $89^\circ \le \theta \le 91^\circ$, and (d) the tumbling motion at $\beta = 1.6$ and $Re = 10$: $T_4 = 5.3$ and $0^\circ \le \theta \le 360^\circ$. The blue solid lines represent the longitudinal displacement X/d, the lateral displacement Y/d, the cosine of the directional angle along x-axis θ, and the vertical coordinates of the nearest point of the surface of the ellipse to the upper wall y^{hi} in different rows, respectively. The yellow dashed lines show the longitudinal velocity V_x/U_{max}, the lateral velocity V_y/U_{max}, and the angular velocity $\Omega d/U_{max}$ in different rows, respectively. The black dashed lines in the last row represent the upper wall. The orientation of the ellipse at different moments is also presented.

If V_x/U_{max} is used as an indicator for the transport efficiency along the flow, the lying motion has the largest value. In addition, the standing motion has a relative larger oscillation in the y position, which explains the large vertical bars on Fig. 4.

D. Influence of the initial configurations on the migration in transit

We further explore how the ellipse starts from rest and goes through a transient migration to reach the final steady state.

We already know from Sec. III B that different initial states determine the type of steady motion in region II, that is, lying or standing motion. Figure 8 exemplifies the transient behavior with different initial configurations for the particle Reynolds number 10 and channel–particle size ratio 1.2. At steady states, the ellipses with initial configurations $(Y_0, \theta_0) = (0.02d, 3^\circ)$ exhibit lying motion, while $(0.2d, 0^\circ)$ exhibits standing motion. For the former, the ellipse always oscillates near the center of the channel with gradually decreasing amplitude during the whole migration, and the θ is always small. For the latter, the ellipse has greater amplitude and reaches the steady state faster. In Appendix C, Figs. 14 and 15 show the schematics of the detailed motion of an elliptic cylinder in the steady state of lying and standing motions, respectively.

For the tumbling type, the initial configurations only affect the transient migration behavior, but do not affect the final equilibrium position as well as the motion type. Figure 9 shows the whole transient behavior with different initial positions and orientations for a Reynolds number of 10 and a channel–particle size ratio of 1.6. For the initial configurations $(Y_0, \theta_0) = (0.2d, 0^\circ)$, the ellipse tumbles from rest and drifts to the equilibrium position $y = 0.27d$. For the initial configuration $(Y_0, \theta_0) = (0.02d, 87^\circ)$, the long axis of the ellipse oscillates around $\theta = 90^\circ$ at the early time, and the CoM crosses the centerline back and forth. The motion is similar to the standing type, with the amplitude of the angle growing and the oscillation of the CoM in the y-direction increasing with time. Eventually, the ellipse moves away from the centerline, drifts toward the focusing position due to inertial lift, and transitions from oscillating to tumbling motion as a result of changes in the velocity gradient. Figure 16 shows the

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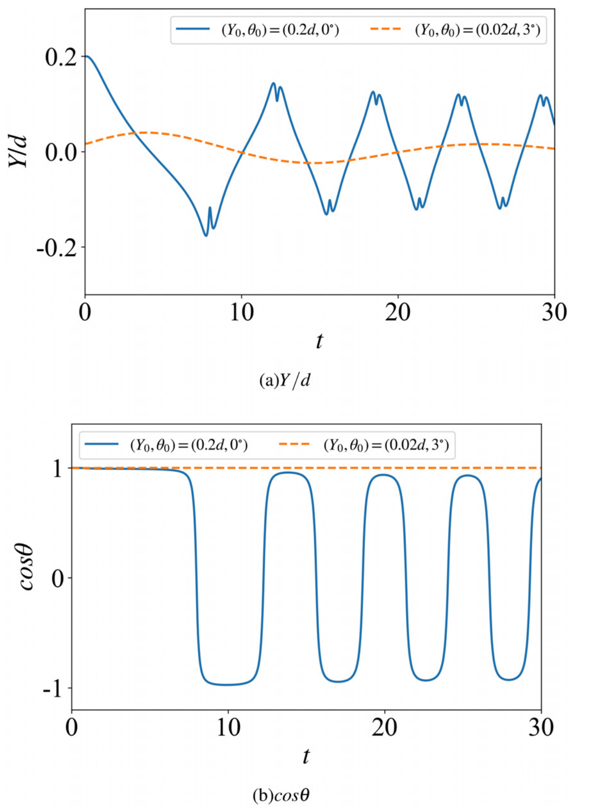

FIG. 8. Transient behavior of the ellipse from rest for $Re = 10$ and $\beta = 1.2$: (a) shows lateral migrations from four different initial configurations, while (b) shows the evolution of cosine of the directional angle of the ellipse. At steady states, $(0.02d, 3^\circ)$ exhibits lying motion, while $(0.2d, 0^\circ)$ exhibits standing motion.

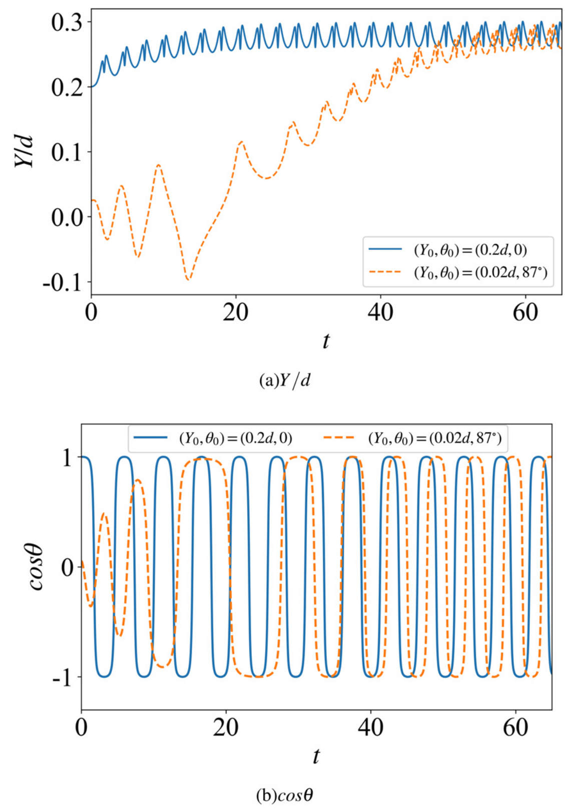

FIG. 9. Transient behavior of the ellipse from rest to steady state of tumbling motion for $Re = 10$ and $\beta = 1.6$: (a) shows positional change of the lateral migration at four different initial positions, while (b) shows the evolution of the directional cosine of the major axis of the ellipse. For clarity, only two initial positions are shown in (b).

schematic diagram of the whole motion of an ellipse in the steady state of tumbling type for $Y_0 = 0.2d, \theta_0 = 0$.

The Reynolds number exerts a significant influence on the migration of an ellipse. Figure 10 illustrates the migration of the lateral displacement of the ellipse over time at varying Reynolds numbers for channel–particle size ratio $\beta = 1.4$ and $\beta = 1.6$. The initial position and orientation are set to $(Y_0, \theta_0) = (0.2d, 0^\circ)$. The simulation time is dimensionless by $t^* = tU_{max}/d$. For an ellipse with channel–particle size ratio $\beta = 1.4$, the ellipse undergoes several tumbles in the transient migration before moving to the center of the channel. The three lines with Reynolds number $Re \leq 1$ are almost coincident. This implies that by decreasing the Reynolds number any further, the particles will still migrate to the centerline. Note that our results for low Reynolds numbers are different from those presented by Seki in his study of Stokes flow.[20] In Seki's results, the ellipse undergoes a periodic tumbling motion for the same initial conditions as ours, with the lateral displacement of the center of the ellipse oscillating periodically at an equilibrium position and not drifting toward the centerline. Nevertheless, even at $Re = 0.1$, our result still exhibit lateral migration, as the inertial forces of the rigid body and the fluid are considered, and the ellipse is always subjected to wall pressure. Reducing the Reynolds number does not eliminate the effect of the wall on the ellipse, but merely increases the simulation time. Assuming that the Reynolds number of our simulation can be infinitely close to zero, our results will be close to Seki's, but never. Furthermore, our simulation results are slightly different from those of Feng and Joseph.[22] They did not observe any tumbling motion in the simulation, and the final steady state is $(Y, \theta) = (0, 0^\circ)$ with no oscillations. In contrast, we observe tumbling motion during the migrations at all Reynolds numbers, and we also find two types of oscillations in the centerline (standing and lying motions). The discrepancy can be attributed to the fact that Feng did not consider the nonlinear effects of the fluid. For an ellipse with a channel–particle size ratio $\beta = 1.6$, the effect of the Reynolds number on ellipse is further magnified. At $Re = 10$, the particle tumbles toward the wall until equilibrium is reached. At $Re <= 1$, the ellipse moves toward the centerline, accompanied by several tumbling motions and ending up with standing motion.

E. Dynamics of a slender elliptical cylinder

The cylinder with $\alpha = 1$ is a special case of elliptic cylinder, and its dynamic behavior has been studied extensively. We now examine the similarities and differences between a slender elliptic cylinder with $\alpha = 3$ and the previous one with $\alpha = 2$ for the same Reynolds number and channel–particle size ratio. We still use a resolution of $\Delta x = b/20$.

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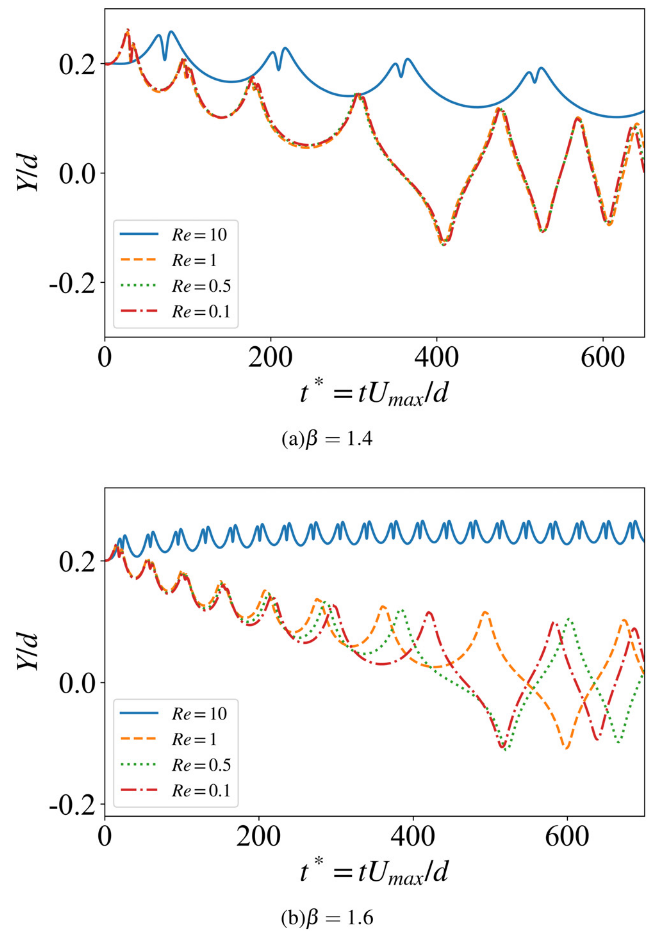

FIG. 10. Comparison of the lateral displacement of the ellipse at different Reynolds numbers.

[Figure 11](#) shows the relationship between the channel–particle size ratio and the equilibrium position at different Reynolds numbers. As with $\alpha = 2$, the equilibrium position of the slender ellipse still remains on the centerline of the channel for $Re = 0.1$ regardless of its initial configurations. As the Reynolds number increases, the equilibrium position of the particle deviates from the centerline for values of β above a certain threshold. Different from $\alpha = 2$, this takes place for $\beta \leq 1.8$ at $Re = 0.5$; $\beta \leq 1.6$ at both $Re = 1$ and $Re = 10$. Another difference is that the curves for $Re = 0.5$, $Re = 1$, and $Re = 10$ with $\alpha = 3$ are closer together than with $\alpha = 2$. For the periodic oscillation in the transverse direction at the equilibrium position, the overall behavior is similar to that of $\alpha = 2$, i.e., the oscillation is more pronounced for a stronger confinement and a larger Reynolds number. In addition, the oscillations are stronger for the standing motion than for the lying motion.

Figure 12 shows a 2D phase diagram about steady behaviors. There are still three types of periodic motion at steady state for the slender ellipse. The type of region I corresponds to that of $\alpha = 2$, while the types in other regions have changed. There are still two types appear in region II, including standing and lying. Compared to $\alpha = 2$,

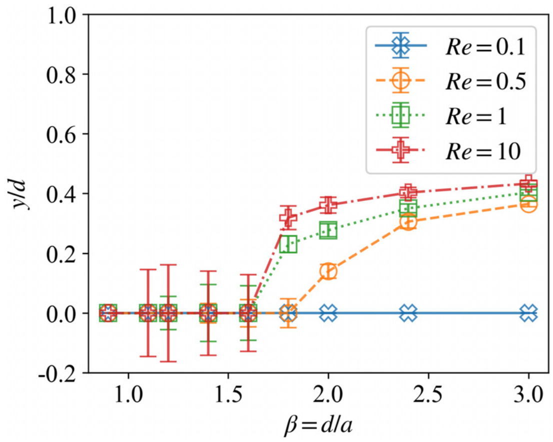

FIG. 11. Focusing positions of the center of mass of the ellipse with $\alpha = 3$ for different channel–particle size ratios β and Reynolds numbers Re at steady states. The vertical bars indicate the amplitude of the position oscillations in the y-direction at steady state.

there is no lying type for $Re = 10$ in this region. The reasons for the multiple types for constituent region II are the same as for $\alpha = 2$. Table II shows the effect of different initial positions and orientations on the types in region II. We can come to the same conclusion as for $\alpha = 2$, that the ellipse is standing or lying in the equilibrium position, depending on particle's initial configurations.

IV. CONCLUSION

In this study, we employ the smoothed particle hydrodynamics (SPH) method, incorporating a novel approach for boundary treatment, to study the dynamics of an elliptical cylinder in confined

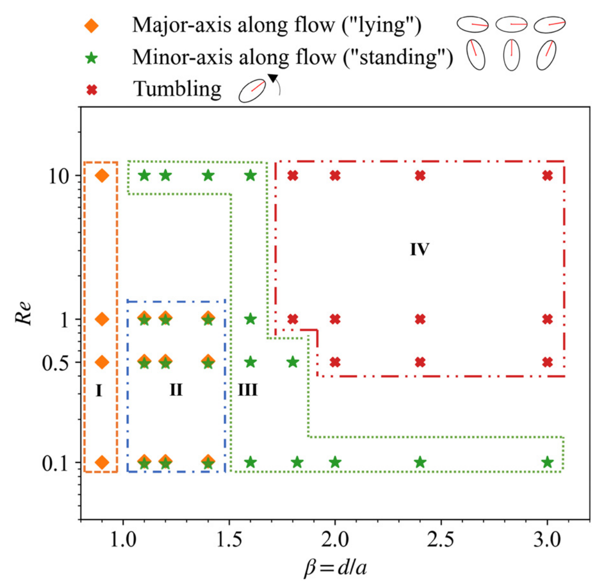

FIG. 12. Phase diagram for the steady behaviors of the ellipse with $\alpha = 3$ with respect to the channel–particle size ratio β and Reynolds number Re.

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TABLE II. Relations between the initial position (Y_0/d)/orientation (θ_0) of the ellipse with $\alpha = 3$ and the steady behavior in region II on Fig. 12. "S" stands for standing motion and "L" for lying motion at steady states. "-" means geometrically not applicable.

Re	β	$Y_0/d\backslash\theta_0$	0°	3°	9°	18°	87°
Re = 0.5	1.1	0.2	L	L	L	S	-
		0.02	L	L	L	S	S
	1.2	0.2	L	L	L	S	-
		0.02	L	L	L	S	S
	1.4	0.2	L	L	S	S	S
		0.02	L	L	S	S	S
Re = 1	1.1	0.2	L	L	L	S	-
		0.02	L	L	L	S	S
	1.2	0.2	S	S	S	S	-
		0.02	L	L	S	S	S
	1.4	0.2	S	S	S	S	S
		0.02	L	L	S	S	S

Poiseuille flow. The simulations are conducted for Reynolds numbers ranging from 0.1 to 10 and channel-to-particle size ratios from 0.9 to 3. Our analysis led to several key findings:

(1) At steady states: within weakly confined channels, the ellipse exhibits a focusing position off the centerline of the channel; within strongly confined channels, the ellipse settles down on the centerline. The boundary between the two types of steady locations is determined by the particle Reynolds number, as summarized in Fig. 4.
(2) Furthermore, at steady states, three distinct types of motions are identified. Focusing positions off the centerline are indicative of inertial focusing phenomena, which promote a tumbling motion. Particles settling down on the centerline are associated with either standing or lying motions, which further depend on the particle Reynolds number, initial orientation/location of the particle, and aspect ratio of the particle.
(3) Two different steady-state trajectories of the center of mass of the ellipse are observed in standing motion: the "M-shaped" trajectory and the "arch-bridge-shaped" trajectory. The M-shaped trajectory resembles to the trajectory observed in steady-state tumbling motion (without flipping over), while the arch-bridge-shaped trajectory is akin to the trajectory associated with steady-state lying motion.
(4) The results at low Reynolds numbers differ from those of Seki[20] because we have taken into account the inertia of both the particles and the fluid. The results also differ slightly from those of Feng,[22] which did not include the non-linear term of the fluid. When the Reynolds number further approaches to 0 but larger than 0, the results are similar to those at $Re = 0.1$, with only longer time scale.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Xinwei Cai: Conceptualization (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Xuejin Li:** Supervision (equal); Writing – review & editing (equal). **Xin Bian:** Conceptualization (equal); Investigation (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

APPENDIX A: SOLID PARTICLE DYNAMICS

In our simulation, the equation for an ellipse in space is represented by

$$\frac{x'^2}{a^2} + \frac{y'^2}{b^2} = 1. \tag{A1}$$

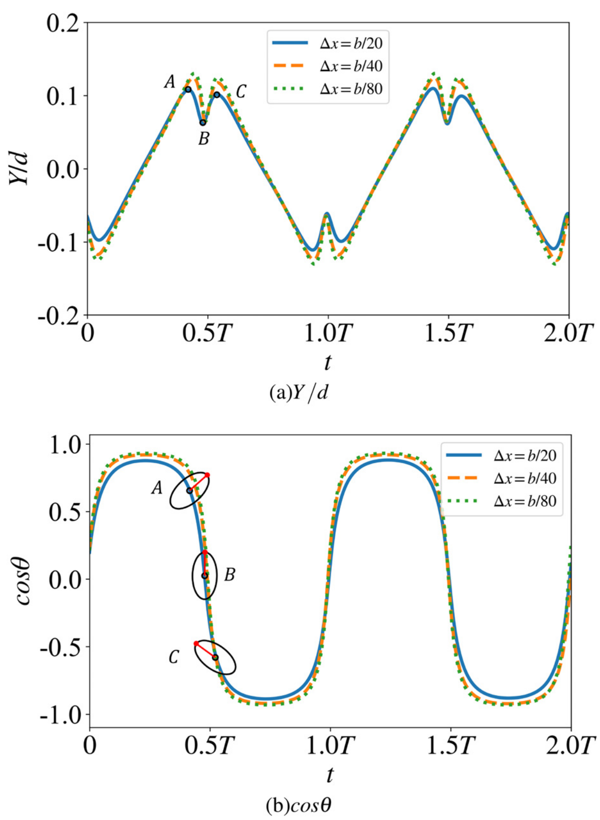

FIG. 13. Resolution study: the lateral displacement (a) and orientation (b) of the ellipse with standing motion type at steady state.

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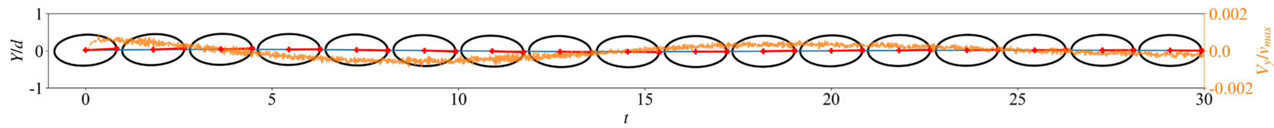

FIG. 14. Motion of an ellipse with aspect ratio $\alpha = 2$ for the steady-state lying type: $\beta = 1.2$ and $Re = 10$. The initial positions of the ellipse are $(Y_0, \theta_0) = (0.02d, 3^\circ)$.

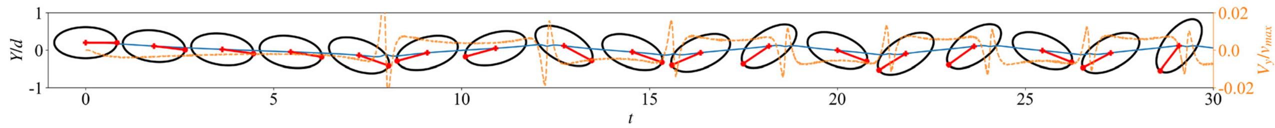

FIG. 15. Motion of an ellipse with aspect ratio $\alpha = 2$ for the steady-state standing type: $\beta = 1.2$ and $Re = 10$. The initial positions of the ellipse are $(Y_0, \theta_0) = (0.2d, 0^\circ)$.

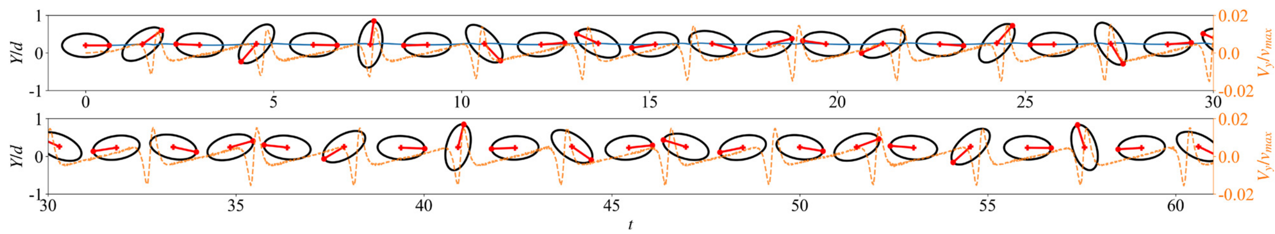

FIG. 16. Motion of an ellipse with aspect ratio $\alpha = 2$ for the steady-state tumbling type: $\beta = 1.6$ and $Re = 10$. The initial positions of the ellipse are $(Y_0, \theta_0) = (0.2d, 0^\circ)$.

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Here, (x', y') represents fixed Cartesian coordinates in space, and a, b are the two semi-axes. The movement and rotation of rigid body can be described by Newton and Euler equations

$$M \frac{d\mathbf{U}(t)}{dt} = \mathbf{F}(t), \tag{A2}$$

$$\mathbf{I} \cdot \frac{d\mathbf{\Omega}(t)}{dt} + \mathbf{\Omega}(t) \times [\mathbf{I} \cdot \mathbf{\Omega}(t)], = \mathbf{T}(t), \tag{A3}$$

where $\mathbf{I}$, $\mathbf{\Omega}$, and $\mathbf{T}$ represent the inertial tensor, the angular velocity, and the torque exerted on the ellipse in the body-fixed coordinate system, respectively. Considering that the ellipse is composed of SPH boundary particles, we use quaternion to update the information of ellipsoid particles

$$\mathbf{q} = \{q_0, q_1, q_2, q_3\}, \tag{A4}$$

with $\sum_i q_i = 1$. The quaternion can represent the rotation matrix $\mathbf{A}(\mathbf{q})$ from a space-fixed $(\mathbf{r})$ to a body-fixed $(\mathbf{r}')$ coordinate

$$\mathbf{r}' = \mathbf{A}(\mathbf{q})\mathbf{r}, \tag{A5}$$

$$\mathbf{r} = \mathbf{A}(\mathbf{q})^T \mathbf{r}'. \tag{A6}$$

At each time step, the resultant force and torque of the rigid body are accumulated by the external force vector of each SPH boundary particle that constitutes the solid. After the force and torque are obtained, the center of mass is updated, and subsequently the boundary particles are updated through the quaternion.

APPENDIX B: RESOLUTION STUDY

A resolution study is conducted for the narrowest channel with a channel-to-particle size ratio β greater than one, aiming to accurately capture the lubricating interactions between the wall and the ellipse. The channel-to-particle size ratio is set at $\beta = 1.1$, and a particle Reynolds number of 10 is selected. The ellipse is standing at steady state, with the shortest distance to the wall among all simulations. The results of three resolutions $\Delta x = b/20$, $b/40$, and $b/80$ are compared in Fig. 13, where the lateral displacements and orientations of the ellipse have consistent behaviors. We choose $\Delta x = b/20$ for most of the simulations to reduce computational cost.

APPENDIX C: MOTION OF AN ELLIPSE AT STEADY STATES

Here, we present the complete motion processes of four steady-state motion types, including three types under the no-slip condition: the lying motion (Fig. 14) with channel-to-particle size ratio of $\beta = 1.2$ and Reynolds number of $Re = 10$; the standing motion (Fig. 15) with channel-to-particle size ratio of $\beta = 1.2$ and Reynolds number of $Re = 10$; and the tumbling motion (Fig. 16) with channel-to-particle size ratio of $\beta = 1.6$ and Reynolds number of $Re = 10$. The horizontal axis represents the displacement in the x-direction of the center of the ellipse, the left vertical axis represents the relative position in the y-direction of the center of the ellipse, and the right vertical axis is the velocity (Vy/v_{max}) in the y-direction and is plotted in the diagram with a yellow dashed line.

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